CONFLOW / CONFORT

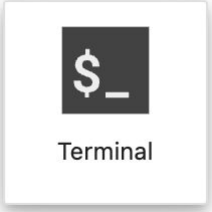
Confort is an updated version of Conflow, an open-source numerical model for flow in eruptive conduits during steady-state pyroclastic eruptions. Confort's improvements include more accurate rheological parameters and equations, evaluations of crystal-bearing rheology, additions of crystal and vesicle size distribution, and integration of degassing in both equilibrium and disequilibrium conditions.

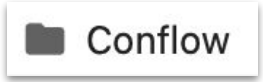
Reference:

Silvia Campagnola; Claudia Romano; Larry G Mastin; Alessandro Vona (2016), "Confort 15 (Conflow improvement)," https://theghub.org/resources/3743.

Mastin, L. G. (2002). Insights into volcanic conduit flow from an open-source numerical model. *Geochemistry, Geophysics, Geosystems*, *3*(7), 1-18.

Activity: CONFLOW on VICTOR

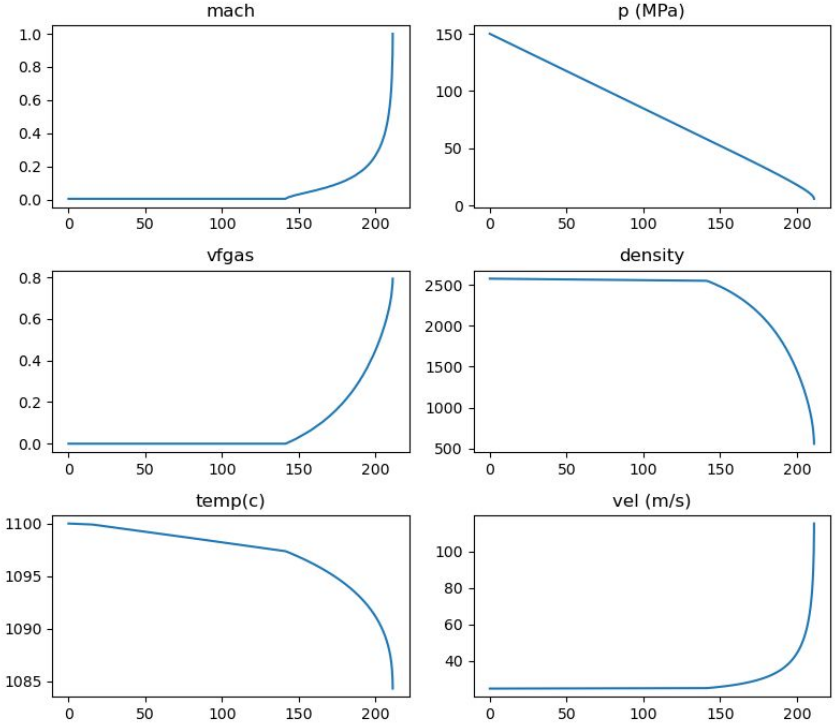


- Login to VICTOR and select a small server size.
- In The Launcher, open the terminal using the icon:
- Type *victor setup* and press enter.
- Select CONFLOW from the list that appears by typing its corresponding number (2) and pressing enter.
- In the home directory (on the left of your screen) the 

folder will load. Double click on the folder to open it.
- Within the Conflow folder, you will see 

open the Notebook by double clicking on it.
- Run the model: 
- Save your visual outputs by holding shift and right clicking on the graph, then selecting “Save Image As…”

Example Outputs:



Change some input parameters

- Modify the composition to reflect ESRP basalts based on the composition used in the viscosity notebook earlier
- Run all the cells
- Open the output figure in a new tab, and compare with the first one. Did anything change substantially? How?
- Now modify the initial pressure (ipa) to a higher value. What happens?
- Keep investigating